Snell's Laws of Reflection and Refraction

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Lecture Objective

By the end of this lecture, you will be prepared to:

- Apply Snell's laws of reflection and refraction (transmission) to solve problems in which plane waves are reflected from and transmitted through the interface between two dielectric media.

Derivation of Snell's Laws of Reflection and Refraction

- Derivation using geometric optics (GO)
- Derivation using boundary conditions arising from Maxwell's equations

Geometric optics can be described as a set of laws obtained from Maxwell's equations in the high frequency limit, that is when all features present are much larger than the wavelength of the electromagnetic signal.

Index of Refraction

- The index of refraction of a medium (n) is defined as the ratio of the phase velocity in free space (c) to the phase velocity in the medium (u_p):

$$n = \frac{c}{u_p} = \sqrt{\frac{\mu\varepsilon}{\mu_0\varepsilon_0}} = \sqrt{\mu_r\varepsilon_r}$$

- Consider two materials with the first having a higher index of refraction then the second. We often say that the first material is “more optically dense” than the second.

Index of Refraction Continued

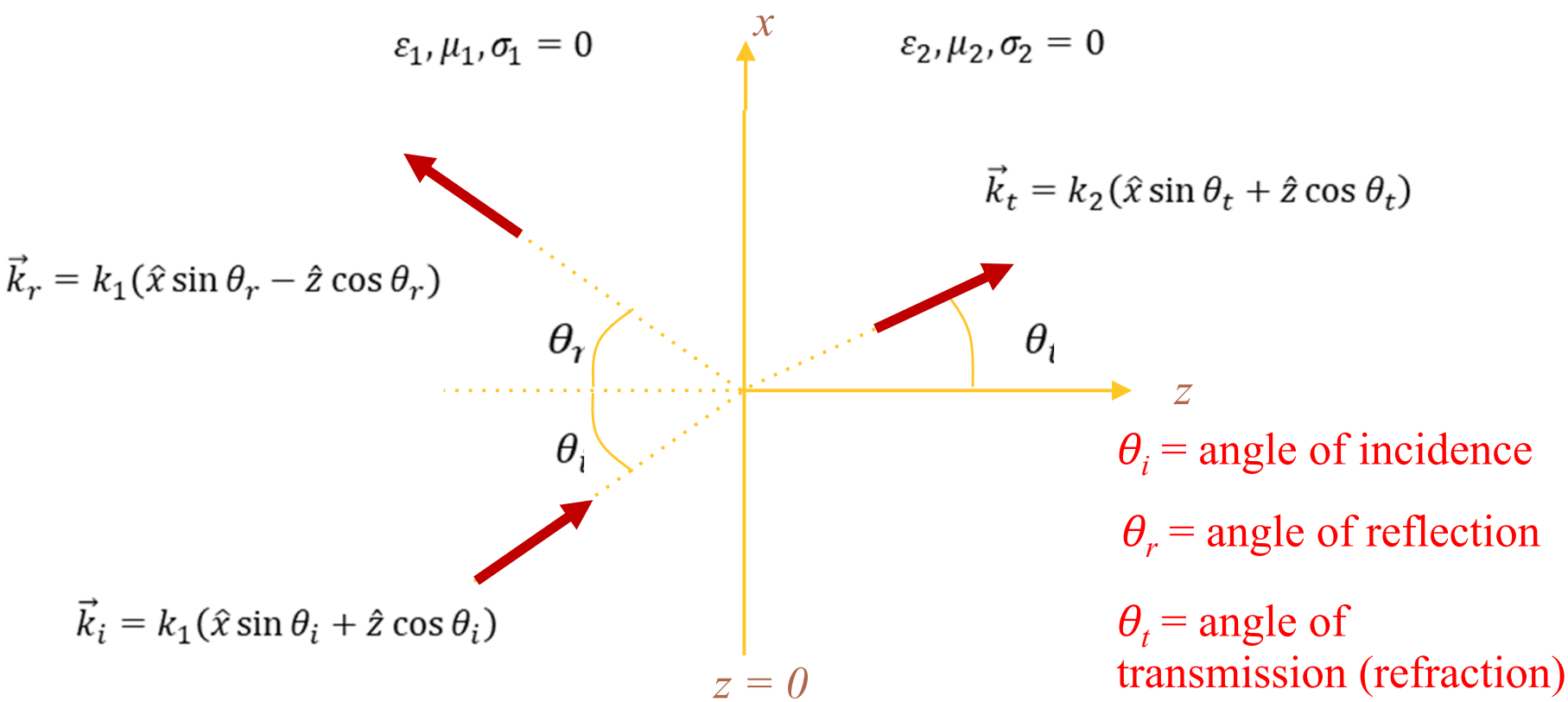
- The wavenumber in the medium can be written in terms of its index of refraction:

$$k = \omega\sqrt{\mu\varepsilon} = \frac{\omega}{c}\sqrt{\mu_r\varepsilon_r} = n\frac{\omega}{c}$$

Question

- Students studying the subject of electromagnetic plane waves sometimes fail to notice that the two quantities denoted by η (intrinsic impedance) and n (index of refraction) are different quantities despite the similar appearance of the two symbols.
- Which equation is the correct equation for calculation of η in a non-magnetic lossless dielectric medium?
- Which equation is the correct equation for calculation of n in a non-magnetic lossless dielectric medium?

Incident, Reflected, and Transmitted (Refracted) Waves



θ_i = angle of incidence

θ_r = angle of reflection

θ_t = angle of transmission (refraction)

GO Derivation

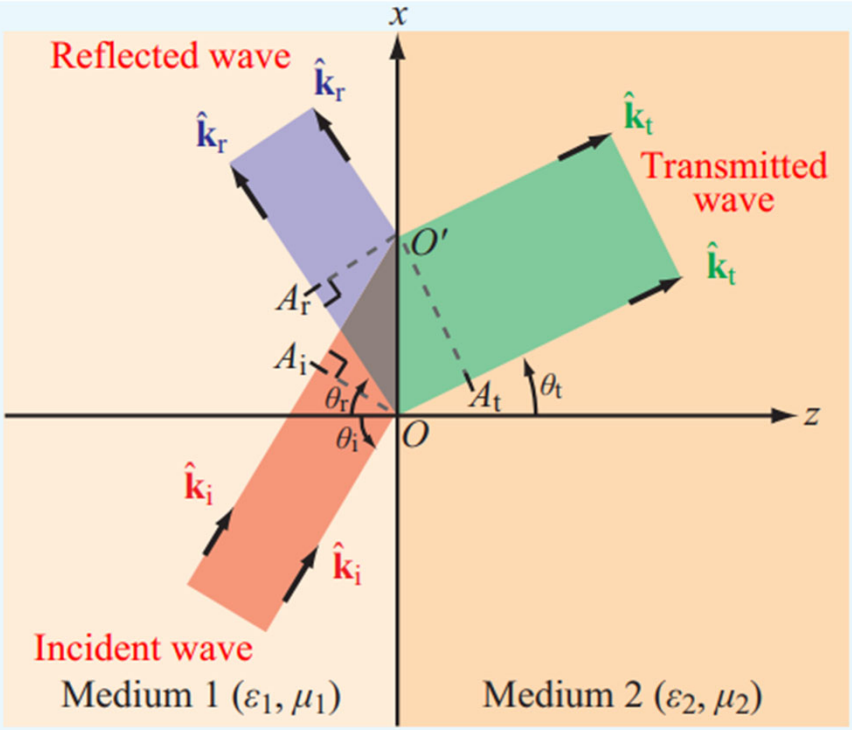


Figure 8-9 Wave reflection and refraction at a planar boundary between different media.

- The dashed lines represent portions of each of the wavefronts (incident, reflected, transmitted).
- The incident and reflected wavefronts move with phase velocity u_{p1}.
- The transmitted wavefront moves with phase velocity u_{p2}.

GO Derivation Continued

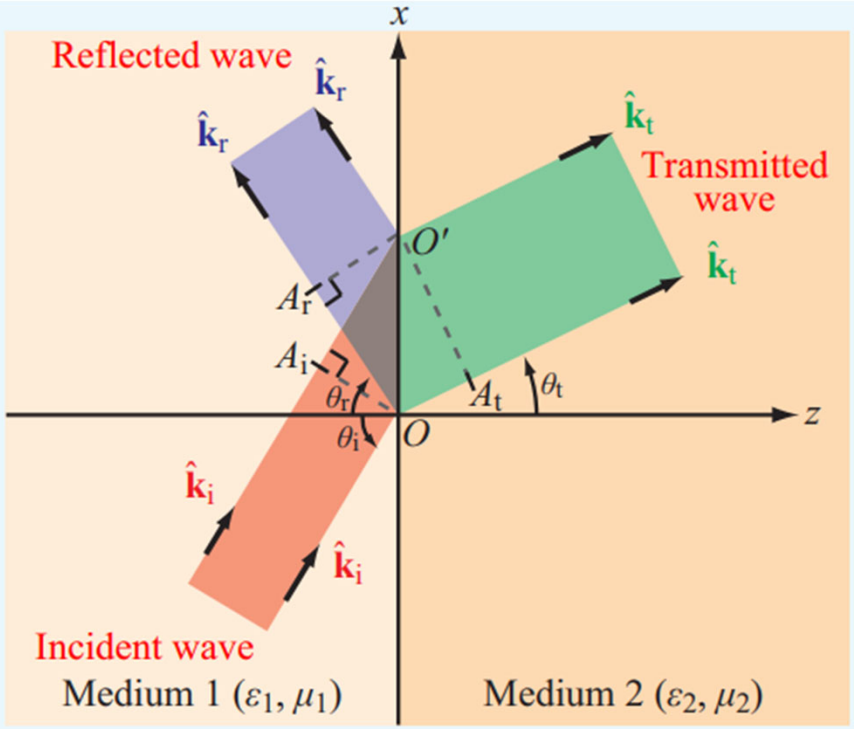


Figure 8-9 Wave reflection and refraction at a planar boundary between different media.

- The time it takes for the incident wave to travel from A_i to O' is the same as the time it takes for the reflected wave to travel from O to A_r and the same as the time it takes for the transmitted wave to travel from O to A_t.

GO Derivation Continued

- Setting each of the phase front travel times equal to the distance traveled divided by phase velocity and to each other, we have

$$\frac{\overline{A_i O'}}{u_{p1}} = \frac{\overline{O A_r}}{u_{p1}} = \frac{\overline{O A_t}}{u_{p2}}$$

- In terms of index of refraction, we have

$$n_1 \overline{A_i O'} = n_1 \overline{O A_r} = n_2 \overline{O A_t}$$

GO Derivation Continued

- From the geometries of each of the three right triangles, we have

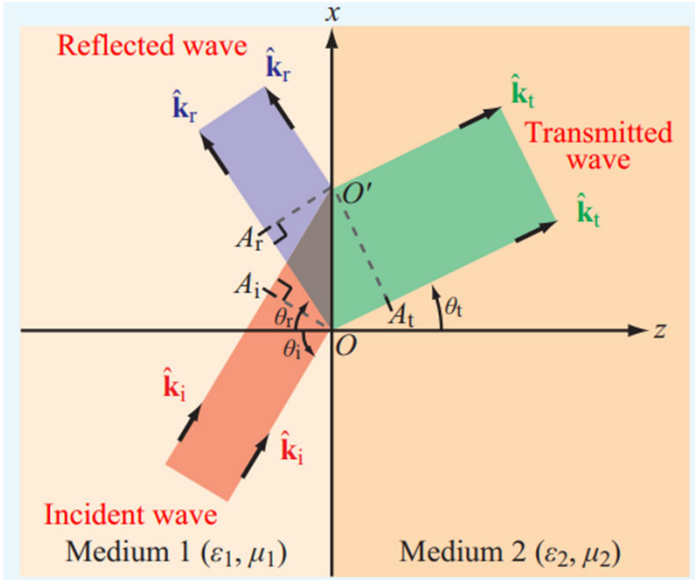


Figure 8-9 Wave reflection and refraction at a planar boundary between different media.

From Triangle $A_i OO'$: $\overline{A_i O'} = \overline{OO'} \sin\theta_i$

From Triangle $A_r OO'$: $\overline{OA_r} = \overline{OO'} \sin\theta_r$

From Triangle $A_t OO'$: $\overline{OA_t} = \overline{OO'} \sin\theta_t$

GO Derivation Continued

- Thus, we have

$$n_1 \sin\theta_i = n_1 \sin\theta_r = n_2 \sin\theta_t$$

- Snell's laws of reflection and refraction are

$$\theta_r = \theta_i$$

$$\theta_t = \sin^{-1}\left(\frac{n_1}{n_2}\sin\theta_i\right)$$

Derivation from BCs (a.k.a. phase matching)

- The electric and magnetic field components of the incident wave are proportional to the propagation factor given by

$$e^{-j\vec{k}_i \bullet \vec{R}} = e^{-jk_1\left(x\sin\theta_i + z\cos\theta_i\right)}$$

- The electric and magnetic field components of the reflected wave are proportional to the propagation factor given by

$$e^{-j\vec{k}_r \bullet \vec{R}} = e^{-jk_1\left(x\sin\theta_i - z\cos\theta_i\right)}$$

Derivation from BCs (a.k.a. phase matching) Continued

- The electric and magnetic field components of the transmitted wave are proportional to the propagation factor" given by

$$e^{-j\vec{k}_t \bullet \vec{R}} = e^{-jk_2\left(x\sin\theta_t + z\cos\theta_t\right)}$$

- Applying the BC at $z = 0$ and enforcing phase matching for the three plane wave components of the field for all values of x yields:

$$k_1 \sin\theta_i = k_1 \sin\theta_r = k_2 \sin\theta_t$$

Derivation from BCs (a.k.a. phase matching) Continued

- Thus, we obtain Snell's laws of reflection and refraction as:

$$\theta_r = \theta_i$$
$$\theta_t = \sin^{-1}\left(\frac{k_1}{k_2}\sin\theta_i\right) = \sin^{-1}\left(\frac{n_1}{n_2}\sin\theta_i\right)$$

Summary

- Snell's laws of reflection and refraction (transmission) can be derived using GO or phase matching in the BCs of Maxwell's equations.
- These laws tell us what happens to an incident plane wave obliquely incident on the boundary between two dielectric half-spaces.

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